

[DC] Emotion Regulation via Eye Gaze

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ABSTRACT

I am a 1st-year PhD student in the field of information science. Currently, there is limited literature catering to emotion regulation using mixed reality embodied agents. The goal of my current research is to design an emotion detection technique that is suited for use with XR embodied agents; intelligent agents that interact with users through a physical or virtual body.

Keywords: Affective AI, eye gaze, emotion regulation, embodied agents.

Index Terms: [Human-centered computing]: Human computer Interaction (HCI) —Agents;

1 INTRODUCTION

With how fast paced everyday life is, controlling anxiety has become a growing concern. The ratio of psychiatric help to patients has been lopsided with additional factors like the COVID-19 barring patients from proper medical support for their anxiety/depression. Moreover, the use of psychiatric drugs for anxiety and depression has been on the increase due to the isolation caused by the pandemic. According to recent reports, antidepressant drug fills increased by 20% after the advent of the pandemic [1].

In order to counter the effect of isolation on this demographic, previous research have already come up with companion robots for stress management: Qoboo (Qoboo.info) and Paro (www.parorobots.com), as shown in Figure 1. Gaming companies have also created serious games that teach the user emotion regulation skills using biosensors [6, 7].

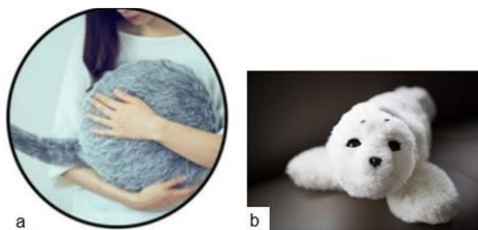


Figure 1: Therapy robots (a) Qoboo; a cat-like robot with interactive tail that reacts to petting (b) Paro; a stress reduction seal robot that reacts to being petted and cared for. (Qoboo.info, www.parorobots.com)

With XR technology both fields can be intertwined. Currently companion robots that manage stress do not measure the user's direct emotional state. I plan to enhance this using existing sensors in the field that are being integrated into XR/VR headsets.

XR can be used to enhance the reaction of a companion robot or

embodied agent to the user. Additionally, methods for emotion regulation relative to an embodied agent have not been analyzed. This paper plans to create an emotion detection/regulation technique specific for embodied/companion agents.

2 EXPECTED CONTRIBUTION

Via designing new emotion detection and regulation techniques that cater to companion agents, we hope to provide a more interactive agent to calm stressed users. By achieving this, we hope to assist users lacking sufficient medical care. We also hope that this method can teach the user emotion regulation skills.

3 LITERATURE REVIEW

Affective AI recently garnered attention since it allows HCI to be relatable by considering the user's emotions. It has been used in so far in the entertainment, automotive and gaming industries [4].

Affective AI usually involves two major steps:

1. Recognizing the user's emotions
2. Reacting to the user's emotions

A particular challenge in affective AI is correct emotion detection. The most common method of training an emotion detector is face recognition via deep learning [5]. Well-recognized companies such as Affectiva [5], have achieved this by acquiring a large dataset to train their models. However, because HMD headsets occlude the user's face, the aforementioned method might not be the most optimal.

Other emotion detection methods have relied on biosensors. For e.g. heart rate variability is often used to approximate the user's stress [6]. Brain Computer Interface (BCI), which relies on the user's neural signals, is also following suite. With BCI technology becoming cheaper, non-invasive ways are now garnering attention [8]. The advent of OpenBCI (openbci.com), has also made it possible to mix BCIs with VR headsets.

In the gaming field, serious games utilizing sensors have previously been used to regulate the user's emotions. One game that was created to regulate the user's fear is Nevermind. Nevermind is a horror game that proves more challenging when it detects a person's fear [6] (See Figure 2). The game achieves this by detecting the player's facial expressions and or heart rate via existing, commonly available, wearable devices or cameras. E.g., Apple watch, Intel RealSense technology [7]. The game operates on a biofeedback system: the game takes the user's biosignals as input and creates a feedback loop to encourage the user to reach a certain emotional goal. During gameplay, the system responds to the user's increased stress by creating scarier scenes [6]. It otherwise rewards the user with calm scenery if they can regulate their emotions. The goal of the game is to train users to exhibit calm emotions when presented with scary situations.

This biofeedback technique however does not use an embodied agent; thus, I want to design an emotion detection technique that caters to VR embodied agents to guide the user.

Another study explored using BCI signals to affect the narrative of a story [8]. The game branches into a specific narrative according to the level of empathy/alpha waves detected: The user is rewarded with a positive narrative when the user shows increased empathy as seen in Figure 3. The technique however suffered from limited affect detection. BCI also caused more

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fatigue in the user. Though the narrative changed according to the user's emotions, the embodied agent used in the narrative had a neutral expression. Additionally, only the color of the embodied agent was changed to express the agent's feelings. We hope to improve on this model by designing an embodied agent that can display emotions.



Figure 2: A screenshot illustrating biofeedback system displayed by the game graphics in case of a calm & stressed person. [6]



Figure 3: Several screenshots from a Serious Game illustrating effects of affective BCI on the embodied agent, when the user succeeds or fails to think about positive thoughts [7]

4 PROTOTYPE DESIGN

My initial prototype design will analyze the user's eye gaze to detect the user's emotions. Previous research has shown that socially anxious individuals avert their gaze from the eyes of individuals showing negative affect in photographs [2]. However, when the studies were conducted with virtual agents displaying neutral facial expressions, participants scarcely reacted. I am currently designing an initial prototype that measures how a virtual agent expressing emotions affects the user's gaze. I am testing heat-maps generated with user's eye gaze to see which areas of the virtual face the user frequents. I also want to track eye gaze patterns in anxious users. My hypothesis is anxious user will avert their gaze from the virtual avatar's face, especially the eyes.

I am designing an experiment where the user is presented with the same avatar displaying a random sequence of emotions during two-minute intervals. The user will be presented with a neutral avatar, a happy avatar, and an avatar displaying negative emotions in random order. The subsequent eye gaze of the user will be measured relative to the avatar's face as shown in figure 4. If the hypothesis is true, I predict that individuals exhibiting anxiety will generate heat-maps where the gaze deviates from the avatar's eyes. As it is difficult to find stressed individuals, I plan to use a set of images known as IAPS (International Affective Picture System) to emulate stress [3] (See Figure 5). IAPS is a set of images that induce emotions in a user. I plan to use IAPS with half the participants to compare stressed and regular eye gaze behavior. To refine the settings and decide the appropriate number of users necessary for the experiment I will run several pilot studies using 5-10 users. The initial tool will be Vive Eye Pro since it has built in eye-tracking and VR capabilities.

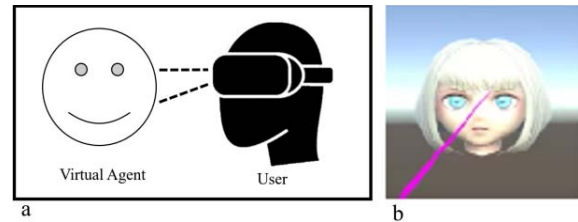


Figure 4: (a) Analyzing user's gaze via heat maps with respect to a virtual agent's face (VR user art by Srinivas Arga; Noun Project). (b) Initial experiment design using the Vive Eye Pro.



Figure 5: A representation of IAPS [3] used for inducing emotions in the user. (www.freepik.es)

5 FUTURE WORK

I plan to improve emotion detection techniques for headset users suited for XR virtual agents and increase the agent's presence to enhance feedback. I hope to teach future user(s) proper emotion management skills via the XR agent. Future use of the Tobii eye tracker to enhance accuracy is under consideration.

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